

Ivar Cashin Eriksson

Curriculum Vitae

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Introduction

Seven years' experience applying machine learning across healthcare, pharmaceutical analytics, manufacturing, e-commerce, and energy. Background in Engineering Physics and Statistical Learning and Data Analytics from KTH. Experience spans probabilistic modelling, autoencoders, sparse Gaussian processes, explainable AI, and production ML. Applying for PhD work on uncertainty quantification for LLMs because the topic aligns with my interest in model evaluation and analysis of model behaviour.

Education

- 2015–2020 **Royal Institute of Technology, KTH – Data Science and Engineering**, GPA: 5.0 (out of 5.0)
Master: Statistical Learning and Data Analytics, Bachelor: Engineering Physics. See thesis.
Relevant coursework: Statistical Machine Learning; Probability Theory; Reinforcement Learning.
- 2017–2022 **Stockholm University, SU – Economics, Supplementary Bachelor's Degree**
- 2019–2019 **Swiss Federal Institute of Technology, ETH – Applied Mathematics, Exchange**

Selected Work Experience

Researcher within Data Science @ RaySearch Laboratories

- Built latent anatomical representations from segmented 3D CT scans using autoencoders.
- Predicted dose-volume histograms and spatial dose distributions with sparse Gaussian processes.
- Redesigned optimisation algorithms to reflect clinical decision hierarchies and validated with clinicians.

Keywords: probabilistic modelling, sparse Gaussian processes, uncertainty-aware evaluation, representation learning, autoencoders, optimisation, medical AI

[Read more...](#)

Data Scientist / ML Engineer @ Signum Life Science

- Work on a forecasting platform for pharmaceutical price, demand, and patient population forecasts.
- Use LLM embeddings for semantic similarity, retrieval evaluation, and matching workflows.

Keywords: forecasting, embeddings, similarity search, Databricks, AWS, XGBoost, ML engineering

[Read more...](#)

Principal Data Scientist @ Valcon

- Built an end-to-end production pipeline for a real-time neural network using NLP, high-dimensional labels and transfer learning.
- Led explainable AI and Git trainings; mentored junior colleagues.
- Presented complex technical topics to non-technical stakeholders.

Keywords: deep learning, NLP, explainable AI, MLOps, mentoring

[Read more...](#)

Skills and Other

- Technology Python, SQL, PyTorch, TensorFlow, Scikit-learn, SHAP, OpenCLIP, Facebook Prophet, MLflow, FastAPI, Docker, MongoDB, Qdrant, Databricks, AWS, MS Azure.
- Methods Statistical learning, probabilistic modelling, sparse Gaussian processes, uncertainty-aware evaluation, autoencoders, CNNs, clustering, explainable AI.
- Language Native English and Swedish; intermediate Danish.

Detailed Work Experience

2024–**Data Scientist / ML Engineer, Signum Life Science, Copenhagen**

Present At Signum, I work across architecture, project management, data science, ML engineering, and software engineering for many ongoing projects. The role requires taking loosely defined business problems and turning them into maintainable technical work.

In my current task, I am using LLMs to embed free text for downstream similarity search and matching workflows. The work focuses on the retrieval layer of RAG-style systems: embedding free text, evaluating semantic similarity, analysing retrieval failures, and improving matching quality in domain-specific workflows.

Alongside this, I am working on a forecasting platform for pharmaceutical price and demand forecasting, including patient population forecasts. I work across architecture, project management, data science, ML engineering, and software engineering for the platform, with Databricks as a core part of the implementation. The platform is being designed to provide thousands of simultaneous forecasts generated by thousands of independently trained models.

Beyond that, I organised an internal week-long hackathon with up to 40 participants.

2022–2024 **Principal Data Scientist, Valcon, Copenhagen**

Led data science projects across pharma, energy, IoT, and manufacturing. Mentored junior colleagues in software architecture and data science best practices, led explainable AI and Git trainings, and regularly presented complex technical topics to non-technical stakeholders.

One project involved training a real-time deep neural network with a custom training pipeline and modular architecture. The training data consisted of natural-language inputs and high-dimensional multi-class labels, which made for a challenging prediction problem. I was responsible for the end-to-end production pipeline.

I was awarded the Valcon People's Choice Award for exemplary work, leadership, team spirit, and scientific curiosity.

2021–2022 **Data Scientist, Valtech, Copenhagen**

Delivered ML solutions in e-commerce. Built time-series forecasts using Facebook Prophet and applied clustering to behavioural data to support personalisation. Worked closely with business stakeholders to ensure actionable outputs, including presentations for C-level stakeholders.

2019–2021 **Researcher within Data Science, RaySearch Laboratories, Stockholm**

At RaySearch Laboratories, I worked on automating radiotherapy planning using deep learning, probabilistic modelling, and multi-objective optimisation. I developed a pipeline that extracted latent anatomical features from segmented 3D CT scans using autoencoders, and used these representations to identify similar patients via a learned similarity metric. Dose-volume histograms and spatial dose distributions were then predicted using sparse Gaussian processes.

Alongside this, I redesigned lexicographic optimisation algorithms to better reflect clinical decision hierarchies. I collaborated closely with medical physicists and clinicians throughout, ensuring the models aligned with oncological standards and practical workflow requirements.

Projects

Project iris End-to-end computer-vision retrieval system for making product images on e-commerce sites interactive. Built object detection, embedding generation, vector indexing, similarity search, backend APIs, persistence, dynamic scraping, and a Chromium extension using YOLOv8, OpenCLIP, Qdrant, FastAPI, and MongoDB.

resume_builder3 Source-grounded agentic workflow for job applications using structured profile data, job listings, LaTeX templates, and GitHub Actions. Produces tailored CVs, cover letters, evidence mappings, and gap reports with human-in-the-loop review.